Tobit Regression

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Introduction

Tobit regression handles outcomes that are observed at the boundary value for cases that fall outside a censoring threshold. Income top-coded at $200,000 (right-censored at the cap), laboratory values below the limit of detection (left-censored at the LOD), drug refill rates that cap at 100 %, all share the same structure: a real underlying value exists, but the observation records the censoring threshold rather than the true value. OLS on the observed outcome is biased toward zero in such cases; Tobit recovers the latent-model coefficients via the appropriate Normal likelihood.

Prerequisites

A working understanding of linear regression, the difference between censoring and truncation, and the Normal distribution as the latent error model.

Theory

For latent $Y_i^* = X_i^\top \beta + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, the observed value is $Y_i = \max(L, Y_i^*)$ for left-censoring or $Y_i = \min(U, Y_i^*)$ for right-censoring. Maximum-likelihood estimation combines two contributions:

$$L(\beta, \sigma) = \prod_{i \notin \text{cens}} \phi\left(\frac{Y_i - X_i^\top \beta}{\sigma}\right) \cdot \prod_{i \in \text{cens}} \Phi\left(\frac{L - X_i^\top \beta}{\sigma}\right).$$

The first product is the standard Normal likelihood for non-censored observations; the second is the probability of censoring for boundary observations. The Tobit model assumes the same coefficients govern both the threshold decision and the observed values; two-part models relax this when those processes differ.

Assumptions

Latent outcome Normally distributed; censoring threshold known and the same for all observations (or known per observation); standard linear-model assumptions on the latent regression.

R Implementation

```r
library(AER)

set.seed(2026)
x <- rnorm(200)
y_star <- 2 + 1.5 * x + rnorm(200)
y <- pmax(y_star, 0)    # left-censored at 0

fit <- tobit(y ~ x, left = 0, right = Inf)
summary(fit)

# Compare with OLS on observed y
coef(lm(y ~ x))
```

Output & Results

`tobit()` returns coefficient estimates on the latent scale, recovering the data-generating parameters. OLS on the same observed (censored) data attenuates the slope toward zero, illustrating the bias from ignoring the censoring structure.

Interpretation

A reporting sentence: "Tobit regression with left-censoring at 0 estimated the latent-scale slope at 1.47 (95 % CI 1.19 to 1.75, $p < 0.001$), close to the true 1.5; OLS on the same data gave 0.98 (95 % CI 0.78 to 1.18), biased toward zero by the unaccounted censoring." Always describe the censoring rule and its threshold in methods.

Practical Tips

- Tobit assumes the same coefficients govern both the censoring and observed values; if these processes differ (some observations are structurally at the boundary, others are observed because of underlying behaviour), use a two-part / hurdle model instead.
- For right-censored time-to-event data, use survival regression (Cox, parametric AFT), not Tobit; the censoring mechanism differs.
- Check Normality of the latent residuals via quantile residuals (`fitdistrplus` for diagnostics); Tobit is sensitive to non-Normal latent errors.
- Tobit is common in economics (income, expenditure); biomedicine more often uses two-part models for detection-limit data because the "below LOD" decision is biologically distinct from the actual concentration.
- For mixed-effects censored outcomes, `survreg(... , dist = "gaussian")` extended with random effects via `survival` workarounds, or `crch` for heteroscedastic censored regression.
- Predictions can target the censored observation (`type = "response"`) or the latent value (`type = "linear"`); choose based on what the question requires.

R Packages Used

`AER::tobit()` for the canonical Tobit implementation; `survival::survreg()` with `Surv(..., type = "left")` as an alternative formulation; `VGAM::vglm()` with `family = tobit()` for richer specifications; `crch` for censored regression with heteroscedastic errors; `pscl::hurdle()` for hurdle alternatives when the threshold decision is distinct from the observed values.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction

- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI